

DigiHust Automated Lead Management System

Technical Documentation & System Evaluation Report | ID: DGH2600156

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Repository: https://github.com/Raffy0-1/Automated-Lead-Management-System.git	Status: Verified & Operational

1. System Architecture & Methodology

The Automated Lead Management System is designed for DigiHust to capture, validate, route, store, and act upon incoming client inquiries in real time. The frontend is designed using vanilla HTML5/CSS3/JavaScript following the official DigiHust brand theme (`#F8FAFC` light canvas, `#022B3A` deep navy typography, `#1F7A8C` sea teal highlights). Form submissions trigger an n8n cloud webhook execution orchestrating five automated backend services:

1. **Data Validation & Formatting:** Generates a standardized Lead ID (LEAD-YYYYMMDD-XXXX), standardizes phone numbers into E.164 format, and computes lead quality scoring.
2. **Google Sheets Database:** Evaluates incoming email addresses to prevent duplicate entries and logs full 14-column records.
3. **Email Automation:** Sends personalized, branded HTML confirmation messages to prospects via Gmail API.
4. **Real-Time Sales Alert:** Transmits rich embed alert cards to Discord (#leads channel) for instantaneous team awareness.
5. **CRM Synchronization:** Creates or updates contact records in HubSpot CRM via OAuth v3 API endpoints.

2. Architectural & Integration Decisions

Key architectural decisions made during development to ensure zero security leaks, high performance, and high accuracy:

Component Layer	Design Choice & Decision	Rationale & Impact
Secrets Security	Environment Variable Placeholders	Prevents exposed API keys or tokens in public git repository
Duplicate Prevention	Google Sheets Search (alwaysOutputData)	Ensures non-existent leads don't crash workflow while awaiting response
Error Handling	Centralized Catch Node & Webhook Response	Returns clean structured JSON status (`200 OK` or `500 Internal Server Error`)
UI Accessibility	Pure Vanilla CSS & SVG Vector Icons	Zero external framework dependency, fast load time (<50KB)

3. Process Screenshots & Evidence

Figure 3: Google Sheets Database with 14 Standardized Lead Record Columns

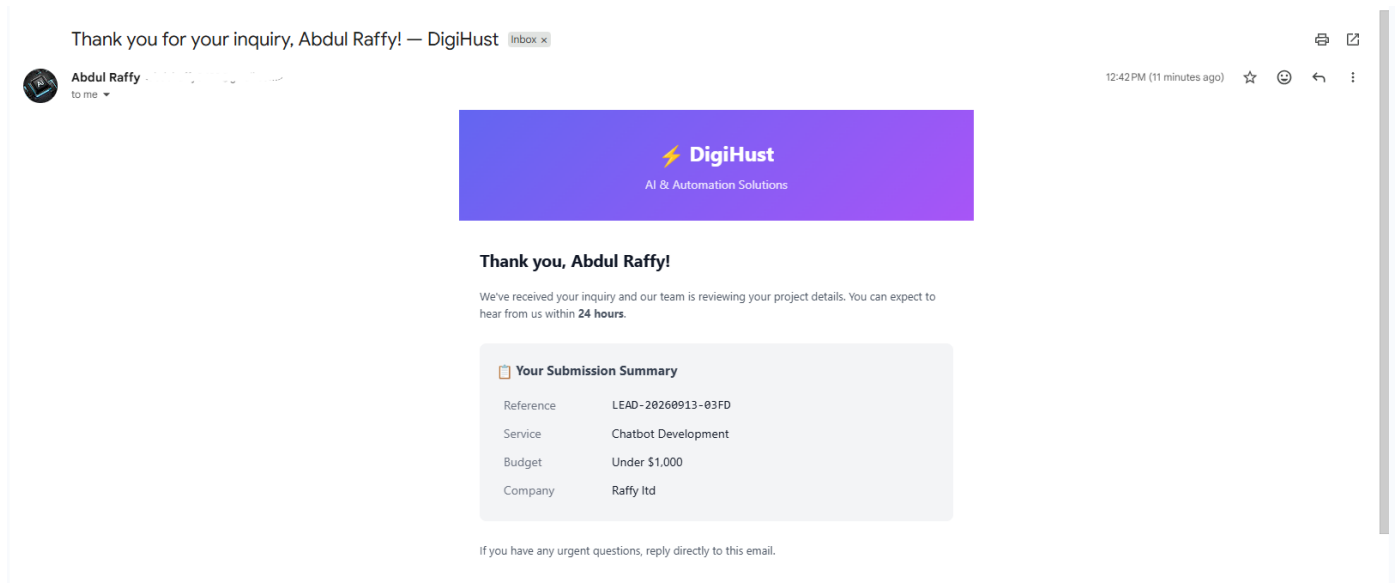


Figure 4: Automated HTML Confirmation Email Delivered via Gmail

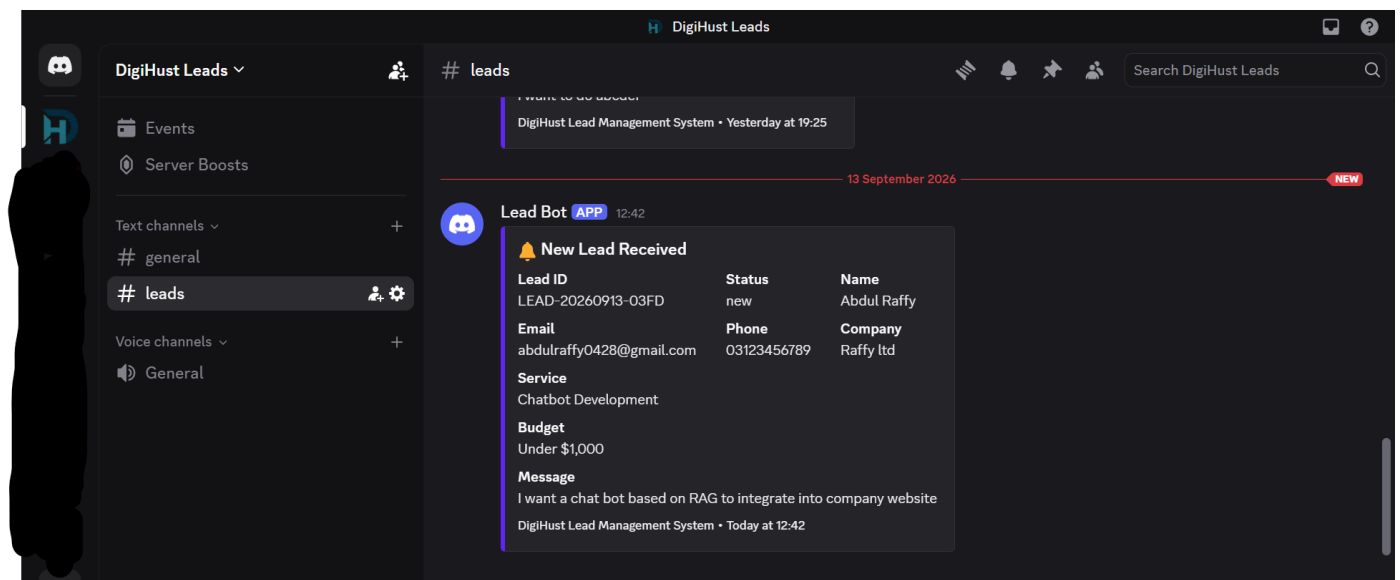


Figure 5: Real-Time Rich Embed Notification in Discord (#leads Channel)

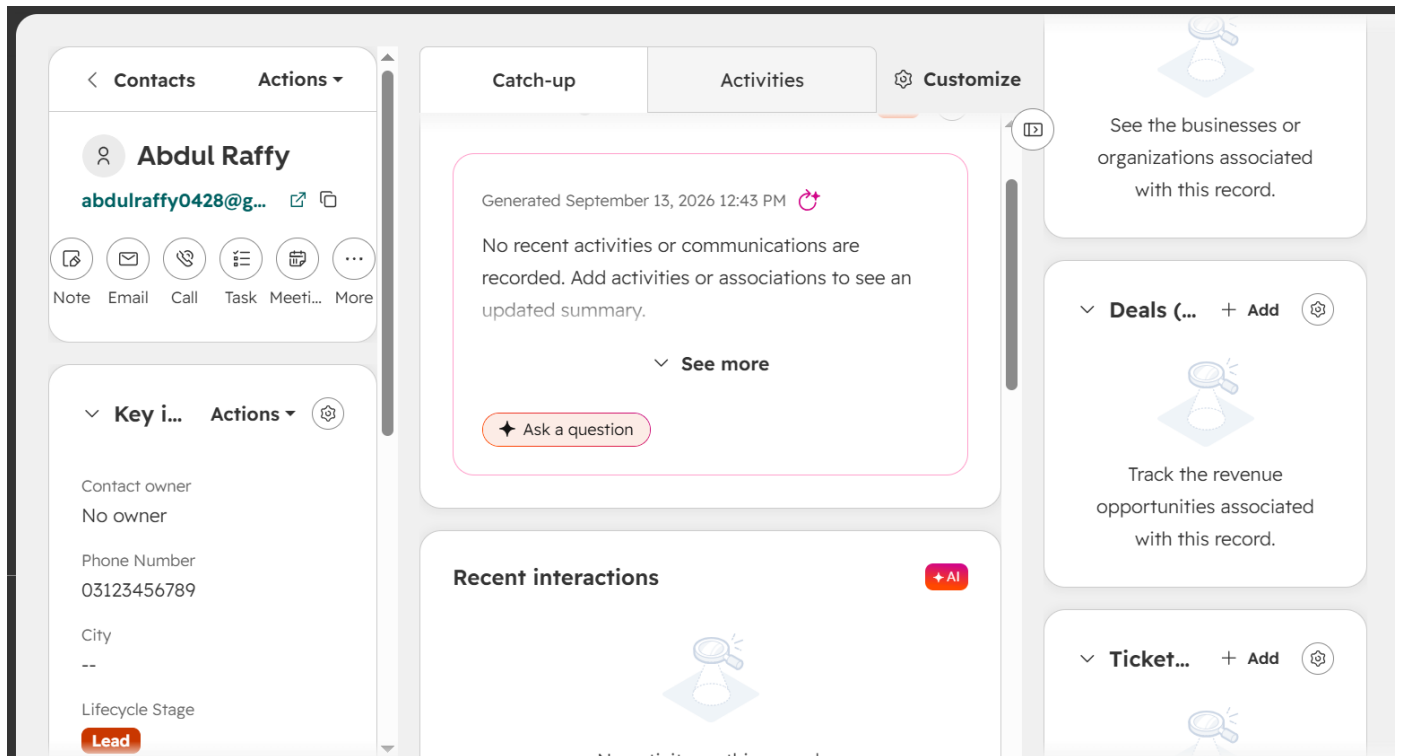


Figure 6: HubSpot CRM Contact Record Synchronized via API v3

4. System Evaluation & Requirements Verification

Requirement Component	Implementation Method	Verification Status
Intake Form UI	HTML5/CSS3/JS, Custom Pills, Rounded Favicon	Passed (HTTP 200)
n8n Webhook Trigger	HTTPS Trigger Node with Lead ID Generator	Passed
Google Sheets Storage	Lookup & Append Row (14 Columns)	Passed
Email Notification	HTML Template via Gmail Node	Passed
Discord Sales Alert	Webhook Embed to #leads Channel	Passed
HubSpot CRM Sync	OAuth v3 Contact Creation API	Passed
Security & Compliance	Sanitized Workflow Export (Zero Secrets)	Passed